Geisinger - Case Length Prediction

Date: 2026-02-16

Executive Summary

This report presents the evaluation of a retrained XGBoost-based machine learning model for predicting surgical case durations at Geisinger Health System. The existing model, previously trained on **OR rooms only**, was retrained on a combined dataset that includes the newly added **procedure rooms**. The primary goals were to (1) extend prediction coverage to procedure rooms and (2) validate that including procedure rooms in training does not degrade the existing OR-room performance.

The model was evaluated using two complementary strategies. First, a **rolling time-aware cross-validation (CV)** with 9 quarterly folds ensures that training data always precedes the validation quarter -- mimicking real-world deployment. Second, **three holdout quarters (2025Q2 -- 2025Q4)** that were never seen during training or hyperparameter tuning provide an unbiased final assessment.

Key Results (Holdout)

Metric	Hospital Est	Model	Improvement
Overall (94,680 cases)			
MAE	22.6 min	19.2 min	3.4 min (15%)
Win-Rate	-	60%	-
OR Rooms (58,556 cases)			
MAE	24.0 min	23.3 min	0.7 min (3%)
Win-Rate	-	51%	-
Procedure Rooms (36,124 cases)			
MAE	20.3 min	12.5 min	7.7 min (38%)
Win-Rate	-	73%	-

OR-room performance is preserved: Including procedure rooms in training did not degrade OR-room results. The model maintains a 0.7 min MAE improvement over hospital estimates for OR rooms (51% win-rate), consistent with prior OR-only model performance.

Procedure rooms provide strong new value: The model achieves a 7.7 min MAE reduction (38%) and 73% win-rate for procedure rooms, where hospital estimates are poorly calibrated for shorter procedures. Overall, the combined model reduces idle time by 29% while keeping delays virtually unchanged.

The sections below describe the data, methodology, and model design, then present all performance results in detail.

Data Sizing Involved in Training

Stage	Count
Raw cases	605,862
After filtering (nulls, invalid timestamps, outliers)	604,399
Rows removed by filtering	1,463
After partial quarter removal (2026Q1 excluded)	593,888

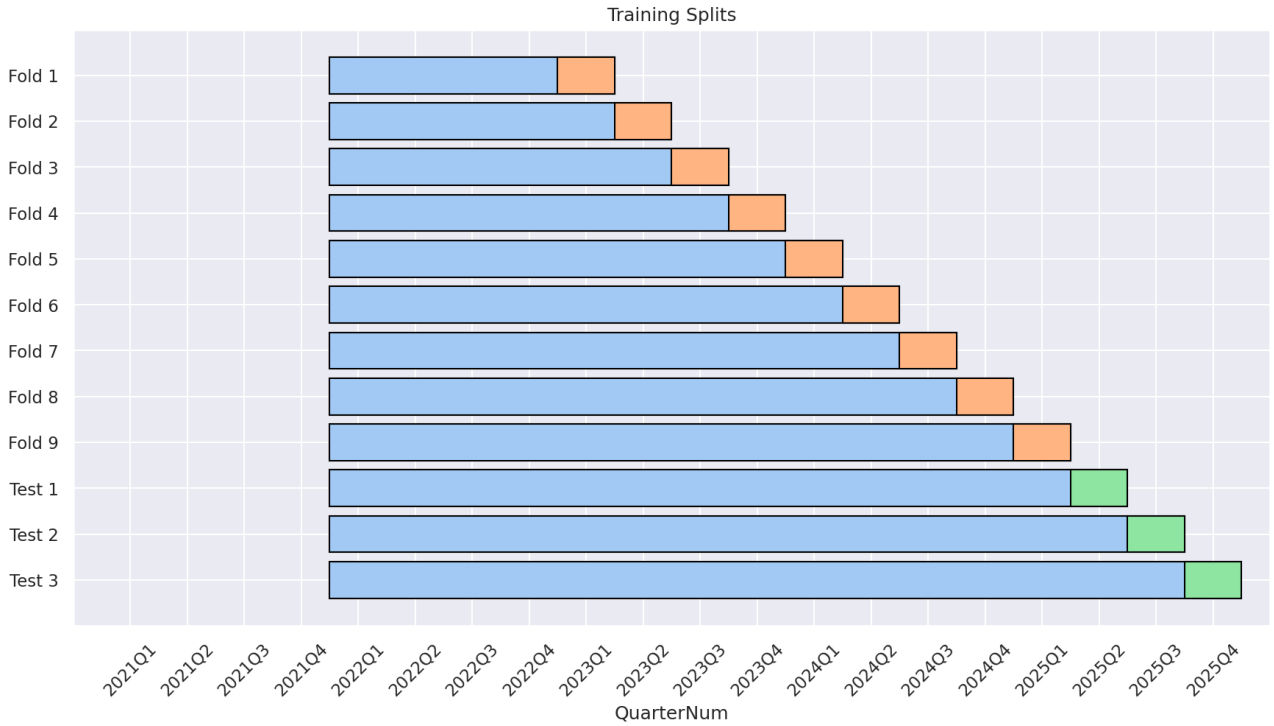
- **Hospital estimate availability:** 100% (all cases have hospital-scheduled durations after filtering)
- **Filtering applied:** target null removal, invalid timestamps, anesthesia-stop outliers (>23h gap between events), discharge outliers

Trained Data Ranges

- Training uses quarterly data starting from **Q1 2022** (all data before 2022 is excluded from training by design).
- Training windows are progressively expanding: each fold trains on all available quarters from 2022Q1 up to the quarter immediately preceding the evaluation quarter.
- **Cross-Validation:** 9 folds, each validating on one quarter from 2023Q1 through 2025Q1.
- The full dataset spans from 2021Q1 to 2026Q1, but only quarters from 2022Q1 onward participate in model training.

Cross-Validation Folds

Validation Quarter	Training Quarters	n_train	n_valid	Hospital Est %
2023Q1	2022Q1 - 2022Q4	113,373	31,862	100%
2023Q2	2022Q1 - 2023Q1	145,245	32,504	100%
2023Q3	2022Q1 - 2023Q2	177,758	30,653	100%
2023Q4	2022Q1 - 2023Q3	208,421	30,398	100%
2024Q1	2022Q1 - 2023Q4	238,831	30,664	100%
2024Q2	2022Q1 - 2024Q1	269,507	31,844	100%
2024Q3	2022Q1 - 2024Q2	301,369	30,192	100%
2024Q4	2022Q1 - 2024Q3	331,572	30,792	100%
2025Q1	2022Q1 - 2024Q4	362,374	25,145	100%



Holdout Data Ranges

- **3 holdout quarters:** 2025Q2, 2025Q3, 2025Q4
- These quarters were **never used** in training or hyperparameter tuning.
- They serve exclusively as the final, unbiased evaluation of model performance.
- Total holdout cases: 94,680

Holdout Splits

Test Quarter	n_train	n_test
2025Q2	387,523	32,072
2025Q3	419,602	31,425
2025Q4	451,031	31,183

Outliers Analysis

- **No explicit minimum duration filter** was applied. The minimum observed duration is 1 min.
- Filtering removed cases with anesthesia-stop timestamps exceeding a 23-hour gap and cases with discharge anomalies. A total of 1,463 cases were removed during this step.
- The duration distribution is **right-skewed**: mean = 80.6 min, median = 54.0 min, 75th percentile = 106.0 min, 90th percentile = 175.0 min, 99th percentile = 392.0 min, max = 1,517 min, std = 80.5 min.
- The model is robust to long-tail cases, as demonstrated by the decile analysis: all deciles show improvement over hospital estimates, including the top decile (175-1,199 min) where model MAE is 64.9 vs hospital MAE of 66.1.

Features Used

671 features total, organized into the following categories:

Category	Description	Count
Patient demographics	Age, sex, smoking status	~18
Case attributes	Number of procedures, number of providers, has-resident indicator	3
Patient class	Case patient class (one-hot)	7
Anesthesia type	Case anesthesia type (one-hot)	13
Urgency level	Case urgency level (one-hot)	15
Surgery patient class	Surgery patient class (one-hot)	7
Admission patient class	Admission patient class (one-hot)	7
Location / Room ID	Room-level one-hot encoding	~170
Parent location / Site ID	Site-level one-hot encoding	~30
Primary procedure laterality	Laterality categories (one-hot)	5
Primary procedure site	Body site categories (one-hot)	53
Primary procedure wound class	Wound classification (one-hot)	6
Surgeon department	Primary department of the surgeon (one-hot)	~200
Surgeon service area	Service area (one-hot)	~10
Surgeon specialty	Primary specialty (one-hot)	~55
Surgeon clinician title	Clinician title (one-hot)	~10
Lag features	Historical trimmed-mean and std by (surgeon + procedure) and by (procedure alone)	4

Note on num_providers: This feature captures the number of providers assigned to the case. It may not be available at scheduling time and is flagged for further investigation.

Surgeon identity: Surgeon durable IDs (raw IDs) are not used directly as one-hot features. Instead, surgeon-specific signal is captured through the **lag features**: historical trimmed-mean and standard deviation by (surgeon + procedure) combination. While this avoids the extreme sparsity of one-hot encoding thousands of surgeon IDs, the lag features still carry surgeon-specific information and could overfit to individual surgeons -- particularly for rare surgeon-procedure combinations with limited history. The trimmed-mean (IQR 0.25-0.75) and rolling window (last 50 cases) provide some regularization, and combinations with insufficient history receive NaN values (handled natively by XGBoost, falling back on other features).

Training Process

- **Model:** XGBoost Regressor with GPU acceleration (CUDA)
- **Objective:** Predict total surgical case duration (in-room to out-of-room) in minutes
- **Missing values:** Handled natively by XGBoost (no imputation required)
- **Categorical encoding:** One-hot encoding for all categorical features
- **Target encoding:** Disabled for this model
- **Lag features:** Rolling trimmed-mean (IQR-filtered, last 50 cases) computed per month using only historical data (no data leakage). Four lag features are computed: 1. Mean duration by (surgeon + procedure) -- last 50

cases, IQR 0.25-0.75 2. Std deviation by (surgeon + procedure) -- last 50 cases, IQR 0.25-0.75 3. Mean duration by (procedure alone) -- last 50 cases, IQR 0.25-0.75 4. Std deviation by (procedure alone) -- last 50 cases, IQR 0.25-0.75

Lag Feature Assignment in Holdout Evaluation

During holdout evaluation, each test quarter receives a **single lag snapshot** corresponding to the last month of training data for that fold. All cases within a quarter use the same lag values, regardless of which month they fall in.

Holdout Quarter	Lag Snapshot From	History Used
2025Q2	March 2025 (month 51)	All cases through Feb 2025
2025Q3	June 2025 (month 54)	All cases through May 2025
2025Q4	September 2025 (month 57)	All cases through Aug 2025

This design mirrors production deployment, where lag features are computed once and used for an entire scheduling period until the next refresh. A case scheduled in April and a case scheduled in June both use the same lag statistics (computed through March). Testing with a single lag snapshot per quarter provides a realistic evaluation of model performance under production conditions, where lags are periodically recalculated rather than updated continuously.

Hyper-Parameter Tuning

- **Method:** Bayesian Optimization (25 iterations + 5 initial random points, early stopping patience = 5)
- **Optimization target:** Minimize MAE on cross-validation folds

Best Parameters Found

Parameter	Value
n_estimators	118
max_depth	11
min_child_weight	16.8
learning_rate	0.072
gamma	0.909
reg_alpha	1.83
reg_lambda	3.04
subsample	0.81
colsample_bytree	0.60
device	cuda

Feature Importance

Feature importance is presented using two complementary methods:

- **SHAP (SHapley Additive exPlanations):** For each case, SHAP assigns every feature a value in **minutes** -- how much that feature pushed the prediction up or down from the baseline (80.6 min, the overall mean duration). The **Avg Impact** column shows the mean absolute SHAP value across all 94,680 holdout cases. This tells you: "on average, this feature moves the prediction by X minutes."
- **Gain:** The feature's share of the total reduction in training loss across all tree splits. This reflects how heavily the model relies on the feature during training, expressed as a percentage.

Top 10 Features

Rank	Feature	SHAP Avg Impact	Effect Direction	Gain %
1	Surgeon + Procedure historical avg duration	21.7 min	Higher avg → longer prediction (e.g., low avg: -27 min, high avg: +39 min)	33.1%
2	Procedure historical avg duration	20.2 min	Higher avg → longer prediction (e.g., low avg: -19 min, high avg: +38 min)	42.7%
3	Number of procedures in the case	5.9 min	More procedures → longer prediction	2.6%
4	Admission patient class: Outpatient	4.3 min	Outpatient: -3.8 min; Non-outpatient: +5.3 min	1.1%
5	Anesthesia type: General	2.3 min	General anesthesia: +1.7 min; Other: -3.7 min	0.6%
6	Procedure historical duration variability (std)	2.3 min	Higher variability → longer prediction	7.8%
7	Number of providers	2.1 min	More providers → longer prediction	1.5%
8	Admission patient class: Inpatient	1.9 min	Inpatient: +3.1 min; Non-inpatient: -1.5 min	0.7%
9	Surgeon + Procedure historical duration variability (std)	1.4 min	Higher variability → longer prediction	1.4%
10	Patient class: Inpatient	0.8 min	Adjusts for inpatient/outpatient interactions	0.5%

Extended Top 11-20

Rank	Feature	SHAP Avg Impact	Effect Direction	Gain %
11	Procedure laterality: Bilateral	0.8 min	Bilateral: +3.4 min	0.2%
12	Patient sex: Female	0.8 min	Male: +0.9 min	0.1%
13	Patient age	0.8 min	Older → slightly longer	0.5%
14	Anesthesia type: MAC-Sedation	0.7 min	MAC-Sedation: -2.0 min	0.2%
15	Procedure wound class: Unspecified	0.7 min	Unspecified: -2.1 min	0.1%
16	Surgeon specialty: Gastroenterology	0.6 min	Gastroenterology: -1.4 min	0.4%
17	Procedure wound class: Clean	0.6 min	Clean: +0.6 min	0.2%
18	Site ID: 300026	0.5 min	Site 300026: +1.8 min	0.1%
19	Patient class: Outpatient	0.5 min	Adjusts for outpatient interactions	0.4%

Rank	Feature	SHAP Avg Impact	Effect Direction	Gain %
20	Has resident	0.5 min	Resident present: +1.1 min	0.1%

Why the rankings differ: Gain ranks procedure-only lag #1 (42.7%) because the model splits on it most frequently — it is available for nearly all cases. SHAP ranks surgeon+procedure lag #1 (21.7 min) because when it is available, it has a larger per-case impact. Similarly, features like "Outpatient" and "Number of procedures" rank higher on SHAP than gain because they have large per-case effects even though the model uses fewer splits on them.

How to read the SHAP values

Consider a case where a **general surgeon** is performing a **laparoscopic cholecystectomy** on an **outpatient**:

1. The model starts at the baseline of **80.6 min** (average case duration).
2. The procedure historical average might be 65 min → SHAP pushes the prediction **down** by ~16 min.
3. This particular surgeon's historical average for this procedure might be 58 min → SHAP pushes it **down** another ~7 min.
4. Outpatient status → pushes it **down** by ~4 min.
5. General anesthesia → pushes it **up** by ~2 min.
6. All other features make small adjustments of less than 1 min each.
7. Final prediction: ~56 min.

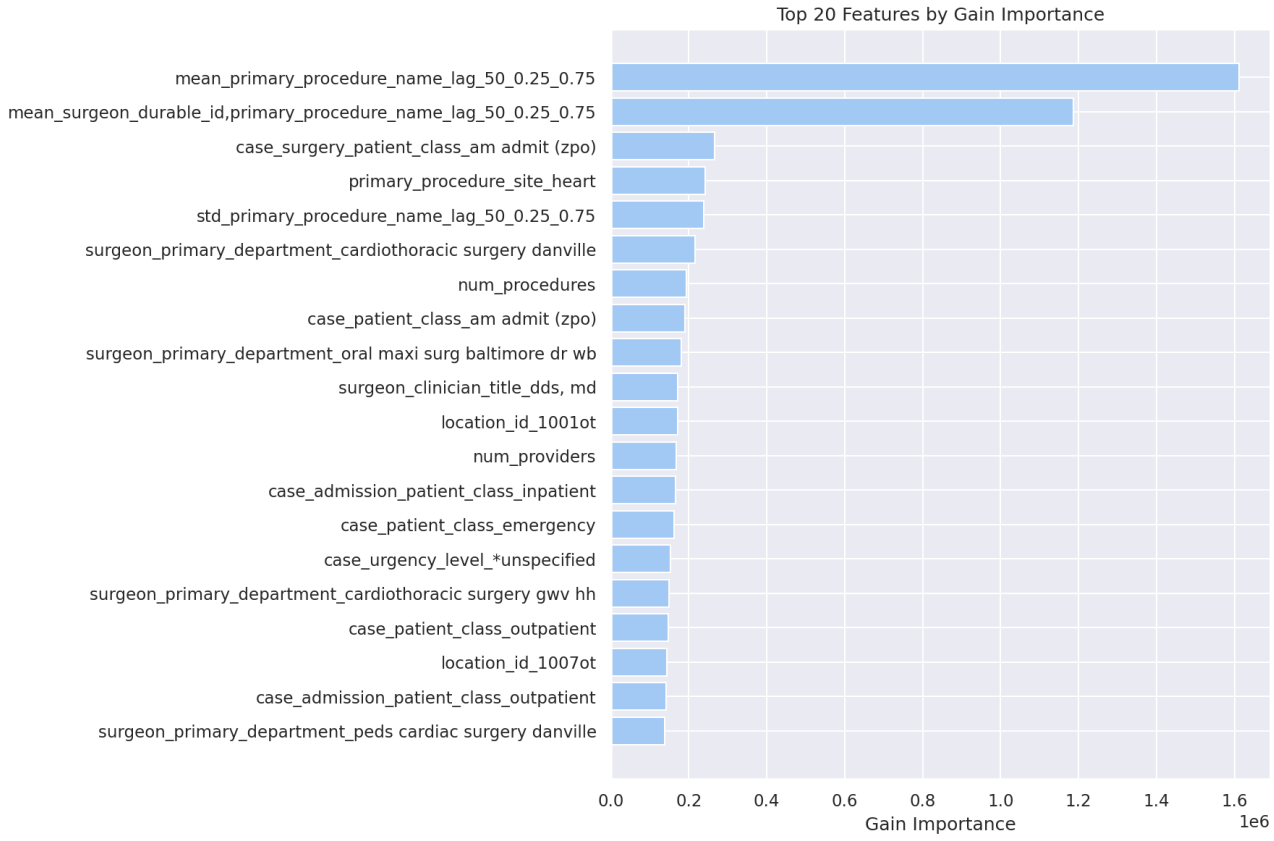
The mean |SHAP| values in the table represent the **average magnitude** of these pushes across all 94,680 holdout cases.

Feature Importance by Room Type (SHAP)

Rank	Feature	OR Rooms (avg impact)	Procedure Rooms (avg impact)
1	Surgeon + Procedure historical avg	23.0 min (26.0%)	19.6 min (30.3%)
2	Procedure historical avg	22.7 min (25.7%)	16.2 min (25.0%)
3	Number of procedures	7.0 min (7.9%)	4.2 min (6.5%)
4	Admission: Outpatient	4.7 min (5.3%)	3.7 min (5.7%)
5	Anesthesia: General	2.5 min (2.8%)	2.1 min (3.3%)
6	Procedure duration variability (std)	2.7 min (3.0%)	1.7 min (2.6%)
7	Number of providers	2.5 min (2.8%)	1.5 min (2.4%)
8	Admission: Inpatient	2.3 min (2.6%)	1.3 min (2.0%)

The feature ranking is consistent across room types. Absolute SHAP values are larger for OR rooms (longer, more variable cases), but relative contributions (%) are comparable.

Key takeaway: The model's predictions are built on a simple, interpretable foundation: - **"How long does this surgeon typically take for this procedure?"** -- 21.7 min average impact - **"How long does this procedure usually take (any surgeon)?"** -- 20.2 min average impact - Together these two features account for **53%** of the total feature impact - **Case complexity** (number of procedures, inpatient/outpatient, anesthesia type) adds ~13 min combined - **Everything else** (demographics, site, specialty, wound class, etc.) adds ~10 min combined



ML Model Measurements - Overall

Metric	Hospital Est	Model	Error Reduction
Cross-Validation			
MAE	22.3 min	19.0 min	3.3 min
MAPE	45%	28%	16 pp
Holdout			
MAE	22.6 min	19.2 min	3.4 min
MAPE	42%	28%	14 pp

Note: All holdout rows have hospital estimates (100% coverage after filtering), so "all rows" is equivalent to "rows with hospital estimates."

By Room Type

Metric	Hospital Est	Model	Error Reduction
OR Rooms (58,556 cases)			
MAE	24.0 min	23.3 min	0.7 min
MAPE	25%	25%	0 pp
Win-Rate	-	51%	-

Metric	Hospital Est	Model	Error Reduction
Procedure Rooms (36,124 cases)			
MAE	20.3 min	12.5 min	7.7 min
MAPE	70%	33%	38 pp
Win-Rate	-	73%	-

OR rooms maintain comparable performance to the prior OR-only model, with a 0.7 min MAE improvement and 51% win-rate -- confirming that adding procedure rooms to training did not degrade OR predictions. **Procedure rooms** are the primary beneficiary of the retrained model, with a 7.7 min MAE reduction (73% win-rate), as hospital estimates are poorly calibrated for shorter procedures.

Cross-Validation Detail by Fold

Fold	Quarter	Model MAE	Hospital MAE	MAE Reduction	Model MAPE	Hospital MAPE
1	2023Q1	19.3	22.6	3.3	30%	48%
2	2023Q2	19.0	22.2	3.3	29%	47%
3	2023Q3	19.1	22.4	3.3	28%	44%
4	2023Q4	19.1	22.5	3.4	28%	45%
5	2024Q1	19.5	22.6	3.0	29%	43%
6	2024Q2	19.0	22.5	3.5	29%	45%
7	2024Q3	18.8	22.5	3.6	28%	44%
8	2024Q4	19.2	22.4	3.2	29%	43%
9	2025Q1	18.2	21.4	3.2	27%	43%
Mean +/- Std		**19.0 +/- 0.4**	**22.3 +/- 0.4**	**3.3**	**28% +/- 1%**	**45% +/- 2%**

Case-Level Performance

- **Overall win-rate (holdout):** 60% of cases where |model_error| < |hospital_error|

Quarter	n_cases	Win-Rate
2025Q2	32,072	60%
2025Q3	31,425	60%
2025Q4	31,183	59%

Win-Rate by Room Type

Room Type	n_cases	Wins	Ties	Losses	Win-Rate
OR Room	58,556	30,095	0	28,461	51%
Procedure Room	36,124	26,415	0	9,709	73%

Win-rate is stable across all three holdout quarters and shows a clear split between room types: procedure rooms achieve a strong 73% win-rate while OR rooms are near parity at 51%.

Error Distribution Comparison

- Model errors are more concentrated around zero compared to hospital estimates.
- Hospital estimates exhibit systematic positive bias (tendency to over-predict case duration).
- Model bias is closer to zero, indicating better calibration overall.
- This is reflected in the idle/delay analysis: hospital idle time is substantially higher than the model's, while delay time is comparable.

Accuracy Within Tolerance (Holdout)

Absolute Tolerance (Minutes) - Overall

Threshold	Hospital Est %	Model %	Improvement
+/- 10 min	40%	50%	+10 pp
+/- 15 min	53%	63%	+10 pp
+/- 20 min	64%	72%	+8 pp
+/- 30 min	78%	83%	+4 pp

Absolute Tolerance (Minutes) - By Room Type

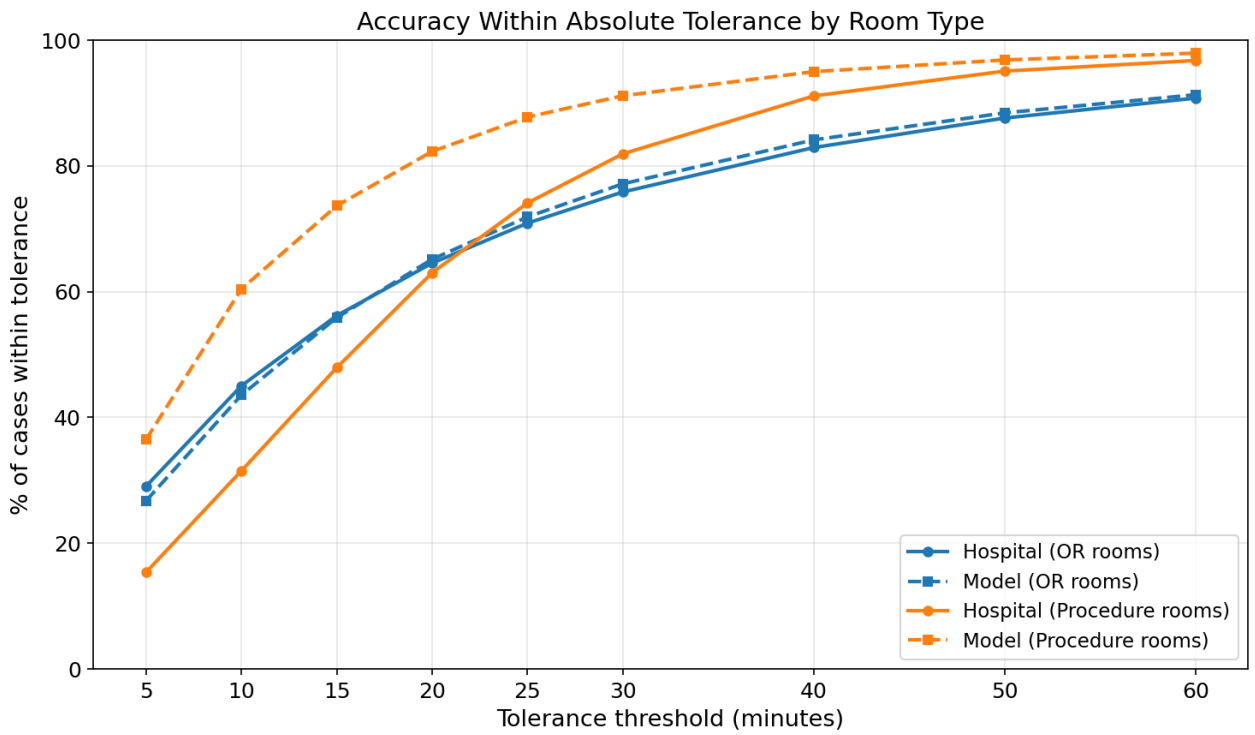
Threshold	OR Hosp %	OR Model %	OR Impr	Proc Hosp %	Proc Model %	Proc Impr
+/- 10 min	45.0%	43.5%	-1.5 pp	31.5%	60.4%	+28.9 pp
+/- 15 min	56.2%	55.9%	-0.3 pp	47.9%	73.7%	+25.8 pp
+/- 20 min	64.5%	65.2%	+0.7 pp	63.0%	82.3%	+19.3 pp
+/- 30 min	75.9%	77.2%	+1.3 pp	81.9%	91.2%	+9.3 pp

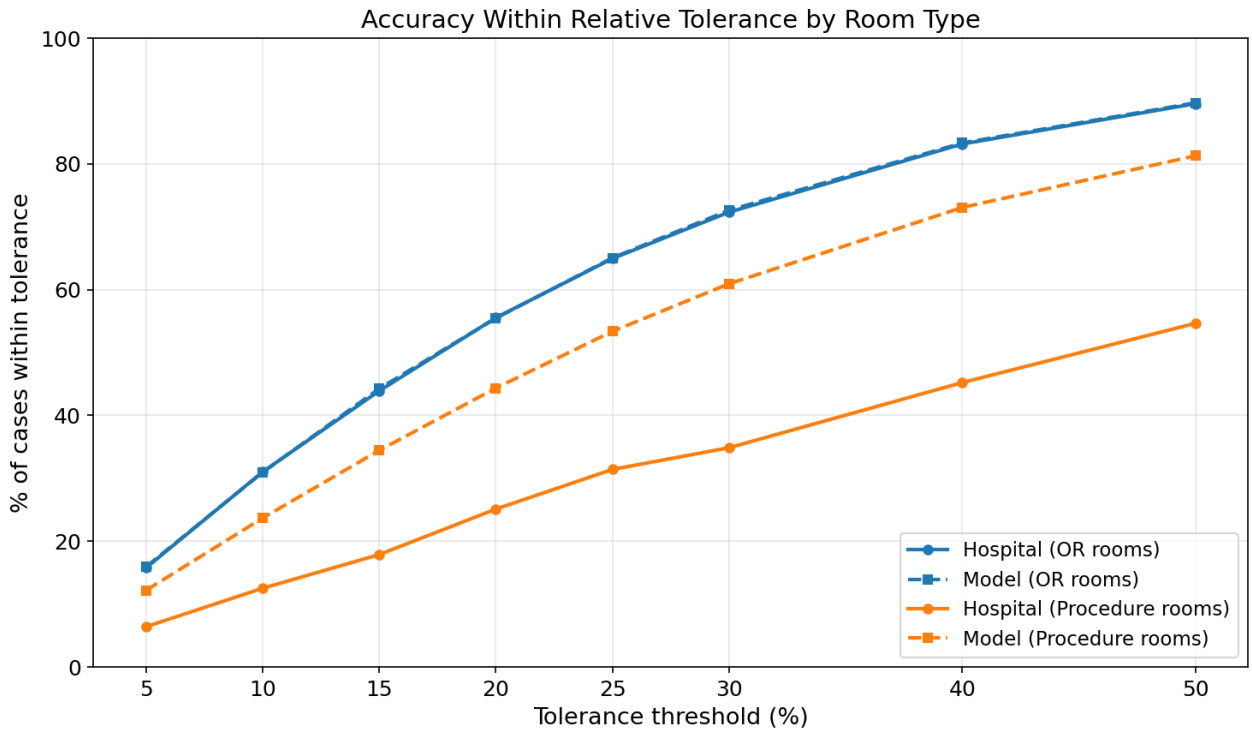
Relative Tolerance (Percentage) - Overall

Threshold	Hospital Est %	Model %	Improvement
+/- 10%	24%	28%	+4 pp
+/- 15%	34%	41%	+6 pp
+/- 20%	44%	51%	+7 pp
+/- 30%	58%	68%	+10 pp

Relative Tolerance (Percentage) - By Room Type

Threshold	OR Hosp %	OR Model %	OR Impr	Proc Hosp %	Proc Model %	Proc Impr
+/- 10%	31.0%	30.9%	-0.1 pp	12.5%	23.7%	+11.2 pp
+/- 15%	43.9%	44.2%	+0.3 pp	17.9%	34.4%	+16.5 pp
+/- 20%	55.5%	55.5%	+0.0 pp	25.1%	44.3%	+19.2 pp
+/- 30%	72.3%	72.6%	+0.3 pp	34.9%	60.9%	+26.0 pp





The model shows dramatically different tolerance profiles by room type. **Procedure rooms** see massive improvements across all tolerance bands (e.g., +28.9 pp at +/-10 min). **OR rooms** show mixed results at absolute tolerances, with slight decreases at tighter bands but improvement at relative tolerances for wider bands. This reflects the model's stronger impact on procedure rooms where hospital estimates are poorly calibrated.

Idle and Delay Analysis (Holdout)

Overall

Metric	Hospital Est	Model	Improvement
Idle (overestimation)			
Total idle minutes	1,125,212	799,718	
Avg idle per case	20.4 min	14.8 min	
Idle reduction	-	-	**29%**
Delay (underestimation)			
Total delay minutes	1,011,353	1,015,650	
Avg delay per case	27.3 min	24.9 min	
Delay change	-	-	**-0.4%**

By Room Type

Metric	OR Hosp	OR Model	OR Impr	Proc Hosp	Proc Model	Proc Impr
Idle						

Metric	OR Hosp	OR Model	OR Impr	Proc Hosp	Proc Model	Proc Impr
Total idle min	601,333	577,639		523,879	222,078	
Avg idle/case	21.7	18.0		19.0	10.2	
Idle reduction			**3.9%**			**57.6%**
Delay						
Total delay min	803,044	785,024		208,309	230,626	
Avg delay/case	27.7	29.7		25.8	16.2	
Delay change			**2.2%**			**-10.7%**

Interpretation: The idle-time reduction is driven almost entirely by the newly added **procedure rooms** (57.6% idle reduction), where hospital estimates chronically over-predict durations. OR rooms see a modest 3.9% idle reduction, consistent with the prior OR-only model. For delay, OR rooms see a slight improvement (2.2% reduction), while procedure rooms see a small increase in delay (-10.7%) -- a trade-off of the much tighter predictions. Overall, extending the model to procedure rooms enables significantly tighter scheduling without materially increasing delays.

By Segment (Holdout)

By Duration Decile - Overall

Decile	Actual Range (min)	n_cases	Hospital MAE	Model MAE	MAE Reduction
0	1 - 18	10,386	15.2	6.6	8.6
1	19 - 25	9,927	12.9	6.6	6.3
2	26 - 32	8,799	13.0	8.2	4.8
3	33 - 41	8,947	13.6	9.6	4.0
4	42 - 54	9,637	14.0	11.7	2.3
5	55 - 70	9,268	16.0	15.1	0.8
6	71 - 92	9,593	19.6	19.0	0.6
7	93 - 121	9,231	23.3	21.8	1.5
8	122 - 174	9,534	32.5	29.3	3.2
9	175 - 1,199	9,358	66.1	64.9	1.3

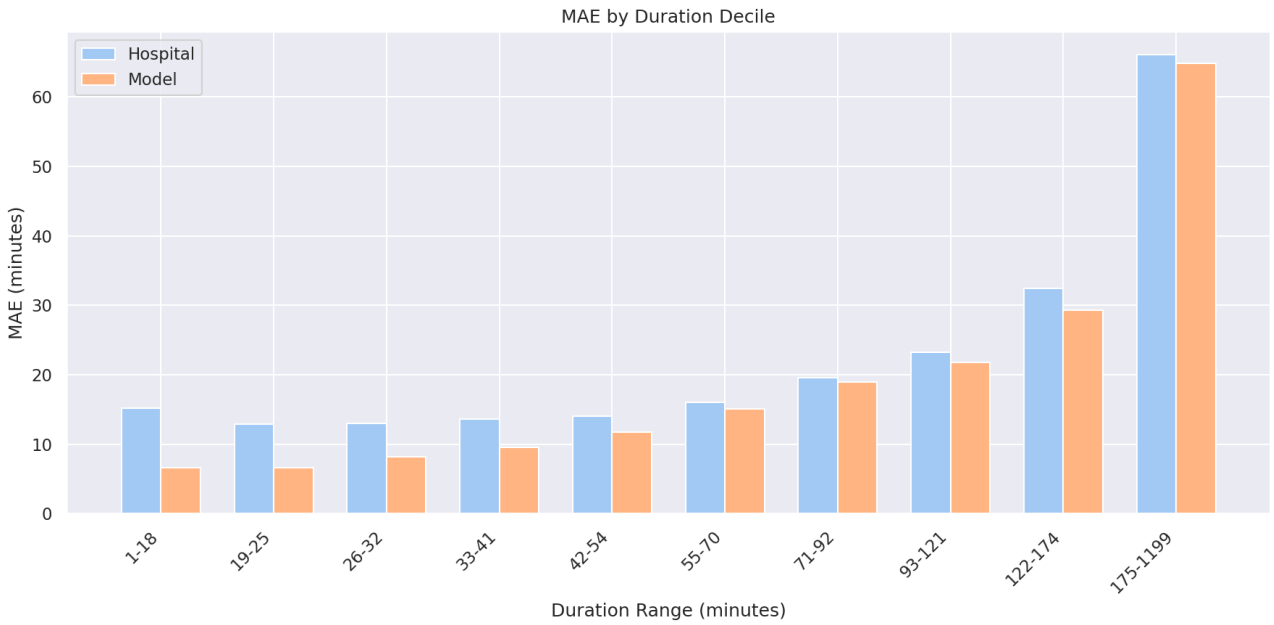
By Duration Decile - OR Rooms

Decile	Actual Range (min)	n_cases	Hospital MAE	Model MAE	MAE Reduction
0	1 - 24	6,357	4.9	5.7	-0.8
1	25 - 35	5,649	8.2	8.5	-0.3
2	36 - 48	5,899	11.8	11.0	0.7
3	49 - 62	5,853	15.1	13.8	1.3
4	63 - 77	5,617	17.6	16.3	1.3

Decile	Actual Range (min)	n_cases	Hospital MAE	Model MAE	MAE Reduction
5	78 - 96	6,036	19.9	19.2	0.7
6	97 - 118	5,671	22.5	20.9	1.6
7	119 - 149	5,812	27.0	25.4	1.5
8	150 - 211	5,812	39.9	36.6	3.2
9	212 - 1,199	5,850	74.2	76.1	-1.9

By Duration Decile - Procedure Rooms

Decile	Actual Range (min)	n_cases	Hospital MAE	Model MAE	MAE Reduction
0	1 - 14	3,714	22.6	7.3	15.2
1	15 - 19	3,945	18.6	7.2	11.4
2	20 - 23	3,869	16.4	6.6	9.8
3	24 - 27	3,552	16.0	7.0	9.0
4	28 - 32	3,533	16.4	8.3	8.1
5	33 - 38	3,272	17.0	9.2	7.8
6	39 - 47	3,492	16.3	10.0	6.2
7	48 - 63	3,591	13.7	12.5	1.2
8	64 - 95	3,581	19.9	20.2	-0.3
9	96 - 804	3,575	45.8	37.7	8.1



Note: The overall model improvement is largest for shorter cases (deciles 0-3). The room-type breakdown reveals why: **procedure rooms** drive the gains, with massive MAE reductions of 9-15 min across all but the longest deciles. **OR rooms** show modest improvements in the mid-range (deciles 2-8) but slight degradation at the extremes (shortest and longest cases), consistent with prior OR-only model performance. The hospital's OR estimates are already well-calibrated for very short and very long OR cases, while procedure room estimates are systematically poor across the board -- this is where the retrained model adds the most value.

By Top Procedures (by volume)

Procedure	n_cases	Hospital MAE	Model MAE	Reduction
Colonoscopy	13,878	13.5	8.1	5.4
EGD Diagnostic	9,218	19.0	10.4	8.6
Cataract Removal w/ IOL	4,177	5.5	5.2	0.2
Coronary Angiography w/ Left Heart Cath	2,835	23.4	17.0	6.3
EGD Ultrasound	1,392	23.2	15.6	7.6
Cystourethroscopy w/ Lithotripsy + Stent	1,287	18.5	17.7	0.9
Coronary Angiography w/ R+L Cath	1,088	16.5	13.1	3.4
Laparoscopic Cholecystectomy	1,048	24.9	23.0	1.8
Cesarean Delivery	999	24.5	16.8	7.7
EGD with Biopsy	912	20.8	7.7	13.1
Hysteroscopy w/ Biopsy/Polypectomy	843	11.0	10.0	1.1
Tonsillectomy + Adenoidectomy (<12)	825	9.9	9.6	0.3
Tympanostomy Tube Insertion (GA)	818	5.2	5.7	-0.5
Injection Spine Lumbar/Sacral	784	2.6	2.3	0.4
Cystourethroscopy w/ Ureteral Stent	768	10.5	9.9	0.6
Catheter Placement Internal Carotid	748	23.3	19.6	3.6
Robotic Total Knee Arthroplasty	742	17.1	15.8	1.3
ERCP Diagnostic	737	28.3	24.6	3.7
Exploratory Laparotomy	715	56.6	54.8	1.8
Laparoscopic Appendectomy	710	19.6	18.8	0.8

Note: The model improves on 19 of the top 20 procedures. The only procedure where the hospital estimate is slightly better is Tympanostomy Tube Insertion (-0.5 min), a very short procedure with already low hospital MAE. The largest gains are seen in EGD with Biopsy (13.1 min reduction), EGD Diagnostic (8.6 min), and Cesarean Delivery (7.7 min).

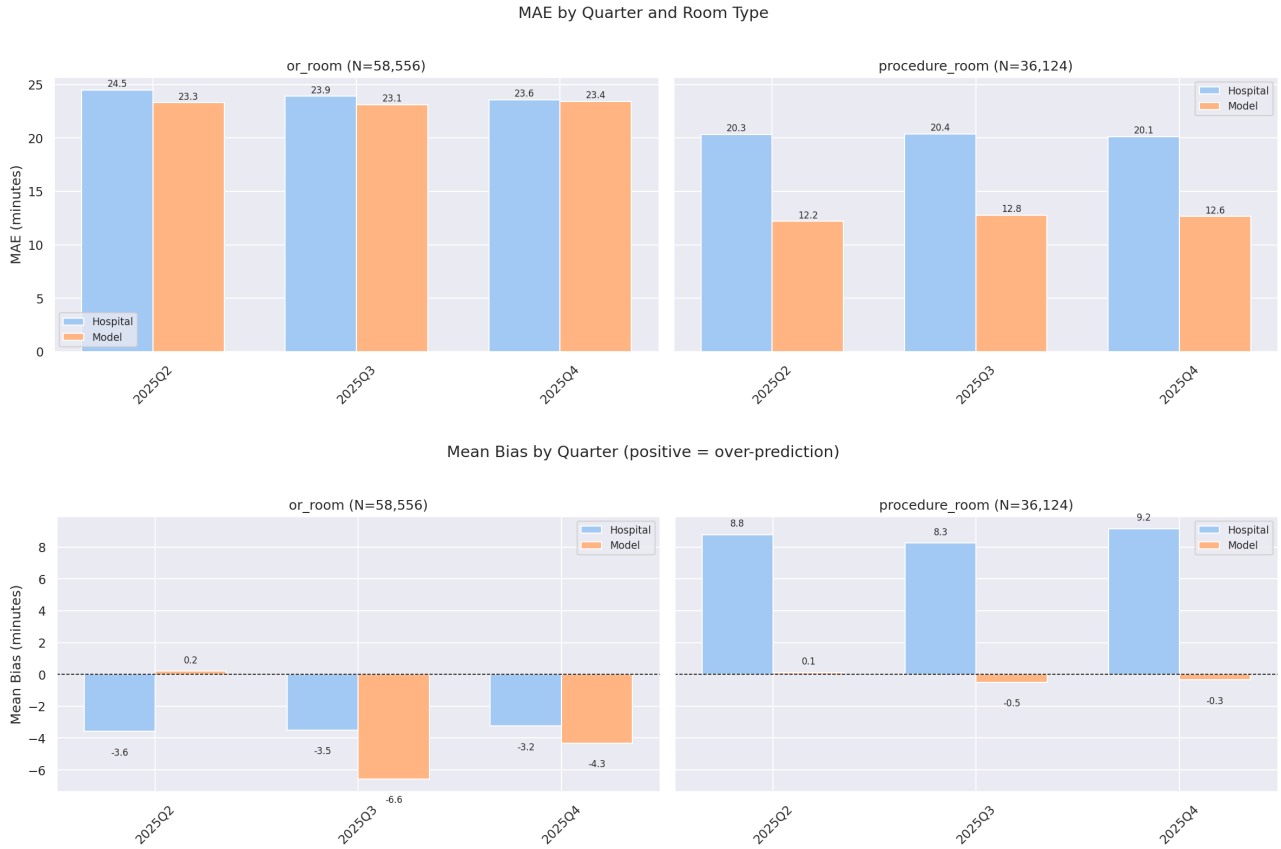
By Room Type

Room Type	n_cases	Hospital MAE	Hospital MAPE	Model MAE	Model MAPE	MAE Reduction	Win-Rate
OR Room	58,556	24.0	25%	23.3	25%	0.7	51%
Procedure Room	36,124	20.3	70%	12.5	33%	7.7	73%

Note: OR-room performance is consistent with the prior OR-only model, confirming no degradation from the combined training. Procedure rooms are where the retrained model adds the most value: hospital MAPE for procedure rooms is very high (70%) because hospital estimates are poorly calibrated for shorter procedures -- the model cuts this nearly in half (33%), achieving a 7.7 min MAE reduction and 73% win-rate.

By Room Type and Quarter

Room Type	Quarter	n_cases	Hospital MAE	Hospital MAPE	Model MAE	Model MAPE
OR Room	2025Q2	19,784	24.5	26%	23.3	26%
OR Room	2025Q3	19,561	23.9	25%	23.1	24%
OR Room	2025Q4	19,211	23.6	24%	23.4	24%
Procedure Room	2025Q2	12,288	20.3	71%	12.2	32%
Procedure Room	2025Q3	11,864	20.4	68%	12.8	33%
Procedure Room	2025Q4	11,972	20.1	71%	12.6	33%



MAE stability: MAE is stable across all three holdout quarters for both room types, varying by less than 0.6 min.

Bias trend: While MAE is stable, the bias plots reveal a pattern worth noting. The model's **overall OR-room bias** shifts from near-zero in 2025Q2 (+0.2 min) to negative in 2025Q3 (-6.6 min) and 2025Q4 (-4.3 min), indicating underprediction. The hospital's bias, by contrast, remains stable across quarters.

Investigation shows this bias shift is **concentrated in long OR cases** (>120 min, ~29% of OR cases). Short cases (<90 min) are unaffected. Procedure rooms show minimal change. The distribution of long cases is nearly identical across quarters (similar counts, percentiles, and means), so the shift is not driven by case mix changes.

Since each holdout quarter is evaluated by a **different model** (trained on progressively more data), the bias shift reflects sensitivity in how different models calibrate long-case predictions. When looking at **long cases only (>120 min)**, the bias is much more pronounced: -18 min for the Q2 model and -31 min for the Q3 model. These long-case figures are far worse than the overall OR-room averages (+0.2 and -6.6) because short cases carry a positive bias that partially offsets the long-case underprediction in the overall average. The hospital also systematically underpredicts long cases (-22 to -24 min across all quarters), confirming this is a challenging segment for any estimator.

This raises a **model stability concern** for the long-case tail: the model's calibration for cases >120 min appears sensitive to the specific training set, while overall MAE remains stable.

Experiments to address OR-room bias: Two additional experiments were conducted to investigate whether alternative model configurations could reduce this bias:

1. **Log-transformed target:** An OR-room-only model trained with log(duration) as the target (predictions converted back via exp()). This did not improve MAE and the bias worsened, as the log transform compresses the long-case tail further, amplifying underprediction for long cases.
2. **OR-room-only model with tuned hyperparameters:** An OR-room-only XGBoost with Optuna hyperparameter tuning (30 trials, 9-fold CV). The tuned model achieved a holdout MAE of 22.94 min vs 23.27 for the all-rooms model -- a modest 0.33 min (1.4%) improvement. However, the bias pattern persisted: overall OR bias remained similar (+0.2, -6.7, -4.2 across Q2-Q4), and per-duration-bucket bias was only marginally improved.

Neither experiment resolved the long-case bias. **A single all-rooms model with linear target remains the preferred approach.** The long-case bias is a fundamental challenge driven by high variance in long surgical cases, not by model architecture or target transformation.

Naive Model Benchmarks

The table below compares the ML model against two naive baselines (simple lag-based predictions using only the historical average, without any ML model) and the hospital's scheduled estimates. All metrics are computed on the **holdout set** (2025Q2-Q4).

Overall

Benchmark	Coverage	MAE (min)	MAPE
Hospital Scheduled Duration	94,680 (100%)	22.6	42%
Naive: Procedure historical avg	93,367 (99%)	23.1	37%
Naive: Surgeon + Procedure historical avg	89,110 (94%)	19.1	27%
ML Model (XGBoost)	**94,680 (100%)**	**19.2**	**28%**

OR Rooms

Benchmark	Coverage	MAE (min)	MAPE
Hospital Scheduled Duration	58,556 (100%)	24.0	25%
Naive: Procedure historical avg	57,343 (98%)	27.6	30%
Naive: Surgeon + Procedure historical avg	53,652 (92%)	23.0	24%
ML Model (XGBoost)	**58,556 (100%)**	**23.3**	**25%**

Procedure Rooms

Benchmark	Coverage	MAE (min)	MAPE
Hospital Scheduled Duration	36,124 (100%)	20.3	70%
Naive: Procedure historical avg	36,024 (100%)	16.0	48%
Naive: Surgeon + Procedure historical avg	35,458 (98%)	13.2	32%
ML Model (XGBoost)	**36,124 (100%)**	**12.5**	**33%**

Key observations:

- **Overall**, the ML model (MAE 19.2) matches the surgeon+procedure naive baseline (19.1) while providing **100% coverage** vs 94%. The 6% of cases without surgeon+procedure history -- typically new surgeons or rare procedure combinations -- are where the ML model adds the most value.
- **OR rooms**: The surgeon+procedure naive (23.0) slightly outperforms the ML model (23.3), suggesting that for OR cases, the historical surgeon-specific average is hard to beat. The ML model's advantage is full coverage. Importantly, these results are consistent with the prior OR-only model, confirming no degradation from combined training.
- **Procedure rooms**: The ML model (12.5) outperforms all baselines, including the surgeon+procedure naive (13.2), indicating that the ML model learns additional signal beyond raw historical averages for shorter procedure room cases. This is the primary new value from the retrained model.
- The **procedure-only naive** baseline is worse than hospital estimates for OR rooms (27.6 vs 24.0) but much better for procedure rooms (16.0 vs 20.3), highlighting how poorly hospital estimates are calibrated for procedure rooms.

Metrics Reported

Metric	Definition
MAE (Mean Absolute Error)	Average of
MAPE (Mean Absolute Percentage Error)	Average of
Win-rate	Percentage of cases where
Error reduction	hospital_error - model_error (positive = model is better)
Idle time	Sum of (predicted - actual) where predicted > actual (over-estimation waste)
Delay time	Sum of (actual - predicted) where actual > predicted (under-estimation causing delays)
pp	Percentage points

Open Topics / Future Work

1. **Investigate removing num_providers feature** if it is not reliably available at scheduling time. This feature ranks 7th in importance (SHAP avg impact 2.1 min) and its removal may modestly reduce accuracy but improve deployment feasibility.
2. **Apply minimum duration filter** (e.g., exclude cases <10 min) to remove potential cancellations or administrative entries that may add noise.
3. ~~**Explore separate models for OR rooms vs procedure rooms.**~~ **Investigated and rejected.** Two experiments were tested: (a) log-transformed target, which worsened bias, and (b) OR-room-only model with tuned hyperparameters, which improved MAE by only 0.33 min (1.4%) without resolving the long-case bias. A single all-rooms model with linear target is preferred.
4. **Consider adding day-of-week and time-of-day features** to capture temporal patterns in case duration (e.g., first case of the day may run longer due to setup).
5. **Add primary procedure code as a feature** via target encoding or embeddings, potentially capturing procedure-level signal beyond what lag features provide.